Chapter # 11
Biostatics and Data Handling

Student Learning Outcomes

After studying this chapter, students will be able to:

- [B-12-K-01] Define biostatistics and its use.
- [B-12-K-02] Define mean, median, mode, standard deviation, range, percentile.
- [B-12-K-03] Calculate mean, median, mode, standard deviation, range, percentile from a given set of data.
- [B-12-K-04] Sketch a bar chart for a given set of data.
- [B-12-K-05] Sketch error bars based off of range or standard deviation for a given set of data on a bar chart.
- [B-12-K-06] Evaluate the appropriate type of figure or chart for a given set of data and/or experiment (bar chart, pie chart, xy axis data figure etc).
- [B-12-K-07] Make the appropriate chart with proper title, labeled axes, legend, axes units.
- [B-12-K-08] Design an appropriate experiment with control group and dependent, independent and control variables.

SLO [B-12-K-01]
Define biostatistics and its use.

BIOSTATISTICS AND ITS USES

Biostatistics is the application of statistical principles and methodologies to the collection, analysis, interpretation, and presentation of data derived from research in the biological, medical, and health sciences. It enables researchers and practitioners to make sense of complex biological phenomena by analyzing data related to living organisms, with the ultimate goal of proposing informed, evidence-based plans or interventions.

Biostatistics finds broad application across a wide range of disciplines including medicine, public health, epidemiology, environmental science, and agriculture. Whether evaluating crop yields, livestock productivity, disease prevalence, or the efficacy of medical treatments, biostatistical tools offer critical insights for effective decision-making and scientific advancement.

Understanding Data and Data Sets in Biostatistics

- **Data**

  In the context of biostatistics, *data* refers to factual information, observations, or measurements related to biological entities. This includes quantifiable metrics such as enzyme activity levels, heart rates, population counts, body weights, or disease incidence rates. These data points serve as the foundation for drawing biological inferences and constructing statistical models.

- **Data Set**

  A *data set* is a systematically organized collection of related data points, typically presented in tabular form for analytical purposes. Each row usually represents an individual experimental unit (such as a subject or sample), while each column corresponds to a specific variable being measured. Every entry reflects a distinct observation.

Example: A table documenting the height and weight of 100 plants grown under varying light or moisture conditions constitutes a data set, facilitating comparative analysis.

Core Components of Biostatistical Research

Effective application of biostatistics involves a sequence of essential steps:

1. **Problem Identification**: The process begins with defining a biological problem—often related to human, animal, or environmental health—that requires scientific inquiry or resolution.
2. **Data Collection and Analysis**: Data are gathered both prior to and following experimental procedures. This data is then rigorously analyzed using statistical tools to identify patterns and draw meaningful conclusions.
3. **Experimental Design**: Biostatistics supports the planning of experiments by determining appropriate sample sizes, durations, variables, and controls necessary to address the research question effectively.
4. **Result Interpretation**: Analyzed data are used to interpret the biological relevance and potential implications of the findings. This step helps in understanding underlying mechanisms or evaluating interventions.
5. **Tool and Policy Development**: Insights gained from biostatistical interpretation often lead to the creation of new diagnostic, therapeutic, or predictive tools, as well as public health policies and scientific recommendations.

Applications of Biostatistics in Biological Sciences

Biostatistics is the application of statistical principles to biological research, enabling scientists to analyze complex data, identify patterns, and make informed decisions. Its applications span agriculture, medicine, genetics, public health, and environmental science.

1. Agricultural and Livestock Demand Forecasting

Biostatistics allows researchers and policymakers to analyze trends in crop yields, livestock production, and dairy outputs in relation to population growth, dietary habits, and climate change. For example, statistical models can predict whether wheat production in Pakistan will meet the projected demand over the next decade or if additional imports are required. Similarly, forecasting models help the dairy industry estimate milk supply fluctuations, enabling better management of feed, breeding schedules, and storage logistics to ensure food security.

2. Evaluation of Medical Instruments and Pharmaceutical Interventions

Clinical trials and controlled studies rely on biostatistical methods to evaluate the safety, efficacy, and optimal dosage of new drugs, vaccines, and medical devices. For instance, during the development of COVID-19 vaccines, biostatistical analyses were essential in determining efficacy rates, side effect frequencies, and optimal dosing intervals. Biostatistics also aids in post-market surveillance, ensuring that medical devices such as pacemakers or insulin pumps continue to meet safety standards.

3. Epidemiological Surveillance and Policy Formation

Biostatistics is critical in tracking disease incidence, prevalence, and transmission patterns in populations. During the COVID-19 pandemic, statistical models predicted infection peaks, hospital bed requirements, and the effectiveness of lockdown measures. Biostatistical evidence guides health policies, vaccination strategies, and resource allocation, supporting both national and global public health planning.

4. Public Health and Population Management

Routine health surveillance relies on biostatistics to assess population health, track birth and mortality rates, and monitor disease burdens. Statistical analyses optimize healthcare delivery by evaluating patient flow, hospital capacity, and treatment outcomes. For example, biostatistical studies can reveal regions with high maternal mortality, guiding targeted interventions, resource allocation, and preventive care initiatives.

5. Genetic Disease Research

Biostatistics plays a central role in understanding the inheritance and prevalence of genetic disorders such as **thalassemia**, **cystic fibrosis**, and **muscular dystrophy**. It allows researchers to calculate carrier frequencies, assess risk factors across populations, and design screening programs. Statistical models also help predict the probability of offspring inheriting genetic traits, informing genetic counseling and preventive healthcare strategies.

6. **Environmental Monitoring and Pollution Control**

Environmental scientists use biostatistics to analyze data on air and water quality, soil contamination, and biodiversity changes. Statistical analyses help identify pollution hotspots, determine sources of contamination, and measure the effectiveness of mitigation strategies. For instance, urban smog trends in Delhi or Karachi can be analyzed to guide emission reduction policies, while water quality monitoring informs river cleanup programs. Biostatistical models also guide reforestation and conservation efforts to restore ecological balance.

7. **Survival Analysis**

Survival analysis is widely used in oncology, cardiology, and other clinical fields to estimate patient life expectancy after treatment. For example, the five-year survival rate of breast cancer patients post-chemotherapy or radiotherapy provides critical insights into treatment efficacy. Survival models also support clinical decision-making, identifying factors that improve outcomes and predicting long-term prognosis for patients with chronic or life-threatening illnesses.

SLO [B-12-K-02]

Define mean, median, mode, standard deviation, range, percentile.

TYPE OF AVERAGES AND THEIR CALCULATION

In everyday life, we frequently summarize complex information using a single representative value. Consider these examples:

- "The average human heart rate is 72 beats per minute."
- "The cancer mortality rate in Pakistan is 25 per 1,000 patients."
- "Wheat yield in Punjab is 2,000 kg per acre."

These numbers are **approximations, not exact figures**. Biological systems are naturally variable: heart rates fluctuate with activity and stress, cancer outcomes differ by region and age, and wheat yields are influenced by soil quality, irrigation, and climate. Yet, these averages are invaluable—they allow us to **see overall patterns, compare populations, and make informed decisions** in medicine, agriculture, and ecology.

In statistics, these representative values are referred to as **measures of central tendency**. They describe the "center" of a dataset—the point around which most observations cluster. The primary measures include:

- **Mean:** The arithmetic average of all data points. It provides a balanced measure of centrality but can be influenced by extreme values.
- **Median:** The middle value when data are arranged in order. It represents the 50th percentile and is less affected by outliers.
- **Mode:** The most frequently occurring value in a dataset, highlighting the most common observation.

While central tendency summarizes a dataset, understanding **variation** is equally crucial. Measures of spread tell us how much data differ from the center:

- **Range:** The difference between the largest and smallest values.
- **Standard Deviation (SD):** A measure of how tightly or widely data cluster around the mean.
- **Percentiles:** Values that indicate a specific position within the dataset, such as the 90th percentile, which is higher than 90% of all observations.

Together, measures of central tendency and variability provide a **comprehensive picture of biological data**. They allow researchers and practitioners to interpret trends, make predictions, and apply this knowledge to real-world challenges, from monitoring population health to optimizing agricultural production.

Types of Averages and Their Calculation

Averages can be calculated using various methods, depending on the nature and distribution of the data. Common measures of central tendency include:

1. **Mean (Arithmetic Average)**

The sum of all data values divided by the number of observations. It is most useful for symmetric distributions without extreme outliers.

2. **Median**

The middle value in an ordered data set. It is a robust measure, less influenced by outliers, making it suitable for skewed data.

3. **Mode**

The value that appears most frequently. Mode is particularly useful in categorical data or distributions with repeated values.

4. **Standard Deviation (SD)**

A measure of the dispersion or spread of data values around the mean. A smaller SD indicates that the data are closely clustered, while a larger SD suggests greater variability.

5. **Range**

The difference between the highest and lowest values in a data set. It provides a basic measure of data spread but is highly sensitive to outliers.

6. **Percentile**

A value below which a given percentage of observations fall. For instance, the 75th percentile (Q3) is the value below which 75% of the data lie. Percentiles are useful in comparing individual scores to a population.

Arithmetic Mean or simply Mean

Definition

The **mean** is a measure of central tendency that represents the arithmetic average of a dataset. It is calculated by summing all the values in the dataset and dividing the total by the number of values. The mean provides a single value that characterizes the overall distribution and is often used to summarize large datasets with one representative figure.

In statistical notation, the sample mean is denoted by $\bar{x}$ (read as "x-bar"). The bar placed over the letter **x** signifies that it is the average of a set of observations.

$$\text{Mean}\left(\bar{x}\right)=\frac{\text{Some of values}}{\text{Total Number of values}}=\frac{\left(\sum x\right)}{(n)}$$

Where:

- $\bar{x}$ = sample mean
- $\sum x$ = sum of all observed values
- n = total number of values in the dataset

Formula:

Direct Method for un-group Data	Direct Method for Group Data
$\bar{x}=\frac{\sum x}{n}$ where, x = Value of data set	$\bar{X}=\frac{\sum fx}{\sum f}$ (where, x = Value of data set and is calculated as $x=\frac{\text{Lower limit}+\text{upper limit}}{2}$ and f = frequency distribution)

Types of Data in Biostatistics

In biostatistics, data are categorized based on how they are organized. Two primary classifications are **ungrouped data** and **grouped data.**

- ### Ungrouped Data

  Ungrouped data consist of raw, individual observations that are not organized into categories or intervals. This form is suitable for small datasets where each value can be listed and analyzed directly.

  Example:

  Individual systolic blood pressure readings from 10 patients: 120, 125, 130, 122, 118, 135, 128, 124, 129, 121

- ### Grouped Data

  Grouped data involve organizing raw data into specific intervals or classes, each with a corresponding frequency. This approach is ideal for large datasets, making them easier to summarize and interpret.

  Example:

  Systolic blood pressure ranges and number of patients:

  - 110–119 mmHg: 2 patients

- 120–129 mmHg: 5 patients
- 130–139 mmHg: 3 patients

Frequency Distribution

A **frequency distribution** is a method of organizing data to show how frequently each value or range of values occurs within a dataset. It may be presented as a table or a graph, such as a histogram. This representation aids in identifying patterns, trends, and outliers in the data and is essential for constructing statistical charts.

Types of Frequency Distributions

1. **Ungrouped Frequency Distribution:**
   - Lists each distinct data value and the number of times it occurs.
   - Suitable for small datasets with limited variation.
2. **Grouped Frequency Distribution:**
   - Organizes data into classes (intervals) and provides the frequency for each class.
   - Suitable for large datasets to simplify interpretation and graphical representation.

Example 1 (Un-group data)

Dataset: Team of world health organization (WHO) planned to assess the prevalence of polio disease in Pakistan. Study to record the polio cases in Pakistan continued for consecutive two years. Number of confirmed Polio patients detected each month of 2022 and 2023 are given in table below. Calculate the mean polio patients per month in each year. Also compare the prevalence of Polio disease in both years.

Sr. no.	Month of sampling	Number of Polio patients in year 2022	Number of Polio patients in year 2023
1.	January	9	7
2.	February	13	9
3.	March	15	13
4.	April	19	17
5.	May	22	18
6.	June	25	17
7.	July	20	15
8.	August	22	14
9.	September	18	11
10.	October	13	9
11.	November	10	8
12.	December	6	6
Total	12 months	192	144

$$\text{Mean number of Polio patients per month in 2022} = \frac{\text{Sum of Polio cases in complete year}}{\text{Total number of Months}} = \frac{192}{12} = 16$$

$$\text{Mean number of Polio patients per month in 2023} = \frac{\text{Sum of Polio cases in complete year}}{\text{Total number of Months}} = \frac{144}{12} = 12$$

Difference of Polio patients per month detected in 2022 and 2023 = 16 – 12 = 04
Prevalence of Polio disease decreased in 2023 compared to 2022 by 04 patients per month.

Example 2 (Group data)

Data set: United Nation-World Food Programme (UN-WFP) conducted a research to assess the impact of food quality on body mass of 100 students of different age groups. Find the arithmetic mean for the *Frequency distribution* of body mass of 100 students as shown in table given below.

Mass of Student Groups Before the Start of Experiment

Sr. No.	Mass (Lbs)	Frequency (f)	$X = \frac{\text{Lower limit} + \text{Upper limit}}{2}$	Frequency *x* mean of limits *(fx)*
1.	35-39	13	$\frac{35+39}{2} = 37$	$13 \times 37 = 481$

2.	40-44	15	$\frac{40+44}{2}=42$	$15\times42=630$
3.	45-49	28	$\frac{45+49}{2}=47$	$28\times47=1316$
4.	50-54	17	$\frac{50+54}{2}=52$	$17\times52=884$
5.	55-59	12	$\frac{55+59}{2}=57$	$12\times57=684$
6.	60-64	10	$\frac{60+64}{2}=62$	$10\times62=620$
7.	65-69	05	$\frac{65+69}{2}=67$	$05\times67=335$
	Total	100		4950

Calculations $=\frac{\sum fx}{\sum f}=\frac{4950}{100}=49.5\text{Lbs}$

49.5 Lbs is the mean mass of 100 students of different groups before the start of experiment

Median

Definition

The **median** is the middle value of a dataset when the values are arranged in ascending or descending order. *It divides the* dataset into two equal halves:

- One half contains values **less than** the median.
- The other half contains values **greater than** the median.

It is denoted by $\tilde{x}$, (read as "x tilde" or "x snake"). It is necessary to arrange the values of dataset in ascending or descending order.

The methods of calculating the median are simple and the value of median is not affected by change in extreme values of dataset.

▶ Median for Un-group Data

- If the number of values is odd, the median is the middle value.

  Median ($\tilde{x}$) = The value of $\left(\frac{n+1}{2}\right)^{th}$ item in the dataset (where n = number of items or data)

- If the number of values is even, the median is the average of the two middle values.

Example 1: (For the dataset with odd number of values)

Dataset: In an experiment to check the effect of specific fertilizer on the growth of the plants, heights of the plants in the experimental dataset was recorded. The heights (in cm) of 13 plants are given below.

58,67,65,70,68,62,63,46,64,49,55,66,72

Find out the Median value of height.

Solution: The median is calculated as follows

The measured heights of plants after arranging in ascending order is as given below:

46,49,55,58,62,63,64,65,66,67,68,70,72

Number of values in the dataset of observed heights = 13

Median ($\tilde{x}$) $=\frac{n+1}{2}=\frac{13+1}{2}=\frac{14}{2}=7$ i.e. value of 7^{th} value in datasheet)

Median ($\tilde{x}$) = 64

Example 1: (For the dataset with odd number of values)

$$\text{Median }(\tilde{x}) = \frac{\left(\frac{n}{2}\text{ th value}\right)+\left(\frac{n}{2}+1\right)}{2}$$

Dataset: The number of fruits produced on each of the 16 plants of experimental group are recorded as below:

3,5,18,21,15,10,8,12,13,7,11,14,9,16,20,24

Find out the median value of fruits produced per plant.

Solution:

The numbers of fruits produced on 16 plants after arranging the values of dataset in ascending order are given below:

3,5,7,8,9,10,11,12,13,14,15,16,18,20,21,24

Total number of values in the above given dataset n = 16 (even number)

$$\text{Median } (\tilde{x}) = \frac{\left(\frac{\pi}{2}\text{th value}\right)+\left(\frac{n}{2}+1\right)}{2} \cdots = \frac{\left(\frac{16}{2}\right)+\left(\frac{16}{2}+1\right)}{2}$$

$$\text{Median } (\tilde{x}) = \frac{(3\text{th value})+(8\text{th value}+1)}{2} = \frac{(12)+(12+1)}{2}$$

$$\text{Median } (\tilde{x}) = \frac{12+13}{2} = \frac{25}{2} = 12.5 \text{ fruits per plant}$$

Activity: (to be solved by the students)

Calculate the median of the following values of data set obtained by measuring the heights of 22 plants in an experimental group to assess the effects of less water availability:

8,9,10,18,10,8,9,7,8,11,23,19,17,15,7,12,14,12,11,14,15,13

Median for Grouped Data

Discontinuous or Unconnected Values in Dataset

To calculate the value of median for discontinuous grouped data, first of all the cumulative frequency (cf) of complete dataset is obtained.

The value of data against $n + 1/2$ the cumulative frequency will be the median for odd number of data.

- Median for dataset with odd number of values will be:

  $\text{Median } (\tilde{x})$ = value of data against $\left(\frac{n+1}{2}\text{th}\right)$ cumulative frequency

- Median for dataset with even number of values will be:

$$\text{Median}(X) = \left(\frac{\text{Class values against } \frac{n^2}{2} + \frac{\pi}{2} + 1^{\text{th}} \text{ cumulative frequencies}}{2}\right)$$

Example for Odd Number Dataset:

Calculate the median of the following dataset obtained by counting the number of flowers on 21 rose plants.

Class (no. of flowers/plant)	1	2	3	4	5
Frequency (no. of plants)	3	5	6	4	3

Calculations:

Class (no. of flowers/plant)	Frequency (no. of plants = n)	Cumulative frequency
1	3	3
2	5	8
3	6	14
4	4	18
5	3	21

Number of plants (n) = 21

$$\text{Median } (\tilde{x}) = \left(\frac{n+1}{2}th\right) \text{ cumulative frequency}$$

$$\text{Median } (\tilde{x}) = \left(\frac{21+1}{2}\right) = \frac{22}{2} = 11$$

Since cumulative frequency 11 is included in 13 which represents the class value 3 , therefore, for cumulative frequency 11 , the class value will be 3 which is the median.

Example for even number dataset

Calculate the median for the following data recorded for height (in cm) of 80 plants.

Class (plant height in cm)	119	120	121	122	123	124	125
Frequency (no. of plants)	4	9	14	18	15	13	7

Calculations:

Calculate the median for the following data recorded for height (in cm) of 80 plants.

Class (plant height in cm)	Frequency (no. of plants = n)	Cumulative frequency
119	4	4
120	9	13
121	14	27
122	18	45
123	15	60
124	13	73
125	7	80

Number of plants (n) = 80

$$\text{Median } (\tilde{x}) = \left(\frac{\text{Class values against } \frac{n}{2}\text{th} + \frac{n}{2} + 1\text{th cumulative frequencies}}{2} \right)$$

$$\text{Median } (\tilde{x}) = \left(\frac{\frac{80}{2}\text{th} + \frac{80}{2} + 1\text{th cumulative frequencies}}{2} \right)$$

$$\text{Median } (\tilde{x}) = \left(\frac{\text{Class values against cumulative frequencies} 40 \,\&\, 41}{2} \right)$$

The class values for cumulative frequencies 40 and 41 are included in the class value of cumulative frequency 45 which is 122 . Therefore, Median $(\tilde{x}) = 122 + 122/2 = 122$.

Advantages of Median

1. **Simple to Calculate and Interpret:** The median is easy to compute, especially in small datasets, and provides a clear indication of the middle value.
2. **Robust to Extreme Values:** Unlike the mean, the median is **not influenced by outliers** or extremely high or low values. This makes it especially useful in **skewed data distributions.**
3. **Applicable in Ordinal and Quantitative Data:** The median can be effectively used with **ordinal data** (data that can be ranked) as well as **quantitative data**, making it versatile in biomedical and statistical analyses.

Limitations of Median

1. **Not Additive Across Multiple Datasets:** The median of multiple datasets cannot be derived by simply calculating the median of each dataset and combining them. This limits its use in cumulative or comparative studies.
2. **May Not Reflect True Central Tendency:** In certain distributions, the median may not lie at the statistical center of the dataset, especially in datasets with **gaps or clusters.**
3. **Limited Use with Discrete and Interval Data:** Median is less effective when dealing with **precise interval measurements** (e.g. age in months, weight in grams), where the exact value is critical for interpretation.

Mode

The **mode** is the value that occurs most frequently in a dataset. It represents the most common observation and is particularly useful in identifying the most typical case in a distribution.

Depending on the data, the mode can take several forms:

- A dataset with **one** mode is called **unimodal.**
- A dataset with **two** modes is called **bimodal.**
- A dataset with **more than two** modes is referred to as **multimodal.**
- If all values occur with the same frequency, the dataset is **amodal** and has **no mode.**

Unlike the mean and median, the mode can be used with both numerical and categorical data, making it a versatile measure of central tendency.

It may be denoted by " $\hat{X}$ "(read as x-hat or x cap).

Mode = the most frequent or repeated value occurring in a given dataset.

Formula for calculating Mode for Group data is given below

$$\text{Mode } (\hat{X}) = \ell + \left[\frac{(f_m - f_1)\times h}{(f_m - f_1)+(f_m - f_2)}\right]$$

Where, ℓ = lowercase boundary of model group

f_m = maximum frequency of a group

f_1 = previous frequency of fm

f_2 = next frequency of fm

h = class interval of model group

▶ Class interval and Modal Group (Modal Class)

1. **Class Interval:** A class interval is a range of values into which data is grouped in a frequency distribution table. Each interval includes a lower and an upper boundary (e.g., 120-129). All intervals are typically of equal width.

   Example:

Blood Pressure (mm/Hg)	Frequency (f)
110 – 119	2
120 – 129	5
130 – 139	3

   Here, 110-119, 120-129, and 130-139 are class intervals.

2. **Modal Group (Modal Class):** The modal group is the class interval with the highest frequency in a grouped frequency distribution. It indicates the most common range of values in the dataset. Using the table above: The class interval 120 – 129 has the highest frequency (5). So, 120-129 is the modal group.

Examples for Calculating MODE

Example #01: Find the mode from the word "BIOLOGY".

Mode $(\hat{X})$ = the most repeated value of the data

Mode $(\hat{X})$ = " O "

Example #02: Find mode from the following data; 3,5,6,6,9,10,9,6,6

Mode $(\hat{X})$ = the most repeated value of the data

Mode $(\hat{X}) = 6$

Example #03: Find the mode from the following data; 3,5,5,6,9,9,10,12

Mode $(\hat{X})$ = the most repeated value of the data

Mode $(\hat{X}) = 5,9$

Example #04: Find the mode from the following frequency distribution;

Groups	Frequency (f)	Class Boundary (CB)	
10-19	5	9.5-19.5	
20-29	25 (f_1)	19.5-29.5	
30-39	40 (f_m)	29.5-39.5	**(model group)**
40-49	20 (f_2)	39.5-49.5	
50-59	10	49.5-59.5	

$$\text{Mode }(\hat{X}) = \ell + \left[\frac{(f_m - f_1)\times h}{(f_m - f_1)+(f_m - f_2)}\right]$$

$l = 29.5, f_m = 40, f_1 = 25, f_2 = 20, h = 10$

$$\text{Mode }(\hat{X}) = 29.5 + \frac{(40-25)\times 10}{(40-25)+(40-20)}$$

$\text{Mode }(\hat{X}) = 33.786$

Standard deviation (SD)

Standard deviation (SD) is a key statistical measure that quantifies the dispersion or spread of data values in a dataset. It indicates how much individual data points deviate from the **mean** (average) of the dataset. In biological research and analysis, understanding variability is critical, and SD offers a clear metric to assess the consistency or variability within a sample or population.

In simple terms, standard deviation answers the question:

"How tightly or loosely are the data values clustered around the mean?"

- When data values are **closely grouped** around the mean, the standard deviation is **small**, indicating **low variability**.
- When data values are **widely spread** from the mean, the standard deviation is **large**, indicating **high variability**.

For example, a biologist might use standard deviation to evaluate:

- The variation in **plant height** in an experimental field,
- The **number of infected individuals** in various cities during a disease outbreak,
- The **fruit yield per plant** in a botanical study.

Interpretation and Symbolism

- For a **population,** standard deviation is denoted by the Greek letter **σ (sigma)**.
- For a **sample**, it is denoted by the Latin letter s.

These symbols represent how far a data point typically lies from the mean:

- A **larger σ or s** indicates greater dispersion.
- A **smaller σ or s** suggests the data points are tightly clustered near the mean.

Relationship to Variance

Standard deviation is directly related to **variance (σ^2)**

Standard deviation is the **square root** of the **variance**.

Although both metrics describe spread, SD is generally preferred because it is expressed in the same units as the original data, making interpretation more intuitive.

Illustrative Example

Consider the quiz results of a biology class:

- If all **40 students** score exactly **77 marks**, the mean is 77, and since every value equals the mean, the **standard deviation is 0**—indicating no variability.
- If a few students score slightly below or above 77 (e.g., 75 or 79), the standard deviation will be **greater than zero** but still **small**, reflecting minor variation.

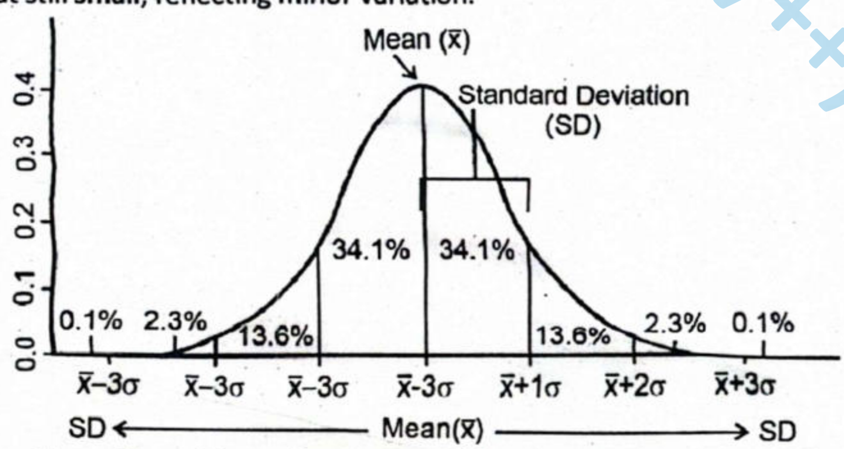

Figure: Normal Curve showing Mean (x) and Standard Deviation (SD)

▶ Calculation of standard deviation

Here are two versions:

- **Population standard deviation** (when the entire population is known)
- **Sample standard deviation** (when only a subset is available)

Calculation Steps:

1. **Calculate the mean** (average) of the data.
2. **Subtract the mean** from each data value to find the deviation.
3. **Square** each deviation to eliminate negatives.
4. **Sum** the squared deviations.
5. Divide:
   - by **n** (number of data points) for population (biased estimate), or
   - by **n – 1** for sample (unbiased estimate).
6. **Take** the **square root** of the result to obtain the standard deviation.

This process provides a reliable indicator of how individual observations differ from the central value, thereby offering valuable insight into data reliability and consistency in biological studies.

Let's calculate the SD for prevalence of COVID in 2021. Number of confirmed COVID cases in four major cities of Pakistan were recorded:

Cities	Number of confirmed COVID cases	Mean of COVID cases
Karachi	90	60
Lahore	80	
Peshawar	40	
Quetta	30	

1. Determine the mean: The mean is 60 .
2. Subtract the mean cases from cases of each city: Karachi = (90 – 60) = 30; Lahore = (80 – 60) = 20; Peshawar = (40 – 60) = –20; Quetta = (30 – 60) = –30;
3. Square the result for each city:
   Karachi $= (30)2 = 900$; Lahore $= (20)2 = 400$; Peshawar $= (-20)2 = 400$; Quetta $= (-30) = 900$
4. Add the results of all cities together: $900 + 400 + 400 + 900 = 2600$
5. Divide this result by the number of scores minus 1 (unbiased), because we are interested in considering these cities as a sample from the entire Pakistan: $2600/3 = 866.7$
6. Determine the square root of this number which is the standard deviation (SD):

The square root of 866.7 = 29.45, Hence, Standard deviation (SD): 29.45

As a general rule large standard deviation means that data is well dispersed away from the mean. A small standard deviation indicates a tight cluster of data points near the mean.

Standard deviation (SD) of a population or sample can be calculated by using following formulas:

Population	Sample
$\sigma = \sqrt{\frac{\Sigma(X-\mu)^2}{N}}$	$S = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}}$
X - The Value in the data distribution	X - The value in the data distribution
μ - The population mean.	$\bar{x}$ - The sample mean
N - Total number of observations	n - Total number of observations

As calculations of SD are complex and there's a high risk of error, so statisticians don't usually calculate standard deviation by hand. We can use a calculator or computer software to find the SD.

Range

Range in Biostatistics

In *biostatistics*, the **range** is a basic measure of dispersion that provides insight into the variability of data collected from experiments or *populations*. It is defined as:

"The difference between the highest and lowest values in a dataset."

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Alternatively, it may be stated as:

"The difference between the maximum and minimum observations."

The resulting value is referred to as the **range of data values** or **range of observations**, and it represents the extent of spread within the dataset. While simple, the range is useful in offering a quick sense of variation among values, particularly in biological studies.

▶ Importance in Biological Studies

In biological research, the range is frequently used to:

- Assess variation within a population.
- Summarise experimental outcomes with respect to spread.
- Identify extreme values or outliers in biological measurements.

▶ Calculation of Range

To calculate the range:

1. **Arrange the dataset** in ascending order (from the smallest to the largest value).
2. **Apply the formula** to determine the range.

▶ Formula for Range

Range = Highest Value – Lowest Value (OR) Range = Maximum Observation – Minimum Observation

Example: Imagine a student studying the heights of a sample of plants in a garden after the treatment of fertilizers. Student collected the following data set of plant heights (in centimetres) from sampled plant population:

1. Raw data Set:
   10 cm,15 cm,18 cm,21 cm, and 25 cm,32 cm,41 cm,28 cm,54 cm,35 cm,26 cm,23 cm,33 cm,38 cm
   40 cm,19 cm,33 cm,43 cm,08 cm,13 cm
2. Arranged data set (ascending order):
   08 cm,10 cm,13 cm,15 cm,18 cm,19 cm,21 cm,23 cm,25 cm,26 cm,28 cm,32 cm,33 cm,33 cm ,
   35 cm,38 cm,40 cm,41 cm,43 cm,54 cm
3. Identify the highest value: 54 cm
4. Identify the lowest value: 08 cm
5. Calculate the range: Range = Highest or Maximum Value - Lowest or Minimum Value
   Range = 54 cm – 08 cm = 46 cm
   The range here is 46 cm, meaning there is a 46 cm difference between the smallest and tallest plants in the sample.

▶ Importance of Range in Biology

In biological research, **range** plays a key role in quantifying the **variability** within a population. It serves as a basic yet informative statistical measure in fields such as **genetics, ecology, and physiology**, where variation among individuals may reflect **environmental influences, genetic diversity**, or **adaptive traits** within a population.

A **small range** indicates that most individuals in the sampled population exhibit **similar responses** to a given treatment or condition. This uniformity implies that the biological response is **consistent** across the population. From a research perspective, a small range is generally considered **favorable**, especially when the **minimum and maximum values both reflect high effectiveness** of the treatment. In such cases, the treatment can be viewed as uniformly successful for the population under study.

Conversely, a **medium range** presents interpretational challenges. The data may not reveal a clear trend or outcome, making it difficult to draw definitive conclusions. In these situations, the researcher may need to **re-evaluate the experimental design, modify the treatment parameters**, or **increase the sample size** to obtain more conclusive results.

A **large *range indicates*** a **high degree of variability** among the individuals in the population. This may suggest that the population includes subgroups with different genetic makeups or environmental exposures, or that the treatment has **inconsistent effects**. While a large range highlights biological diversity, it may complicate data interpretation, as the treatment appears to be effective for some individuals and ineffective for others.

Limitations of Range

Despite its usefulness, range has **notable limitations** that must be considered when interpreting biological data or planning further research. The **range only considers the extreme values**—the maximum and minimum—and **ignores all intermediate data points**. This makes it a **sensitive statistic** that can be **skewed by outliers.**

For example, if nearly all the plants in a treatment group exhibit **high productivity**, but **only one or two** show **very low productivity**, the range will be large. This might misleadingly suggest high variability, even though **most individuals responded similarly**. Therefore, **relying solely on the range** without considering the **distribution of the entire dataset** can lead to inaccurate conclusions.

To address this limitation, researchers often use range in combination with **other statistical measures** such as **standard deviation, interquartile range**, or **variance** to obtain a more comprehensive understanding of data spread and consistency.

Percentile

A **percentile** is a statistical value that indicates the **percentage of data points in a dataset that fall below a particular value.** For example, the **25th percentile** represents the value below which **25% of the data points** lie. Percentiles are widely used in biological sciences to **help** interpret large datasets and provide meaningful comparisons across populations. Common applications include **test scores, growth measurements**, and other **biometric indicators.**

Percentiles divide an ordered dataset into **100 equal parts**, with each percentile representing one-hundredth of the data distribution. It is possible to identify a percentile at any value between **0 and 100**, such as the **99th percentile, 44th percentile**, or **9th percentile.** A percentile score refers to the **value or data point in the ordered dataset** that indicates the proportion of observations that lie **at or below** that value.

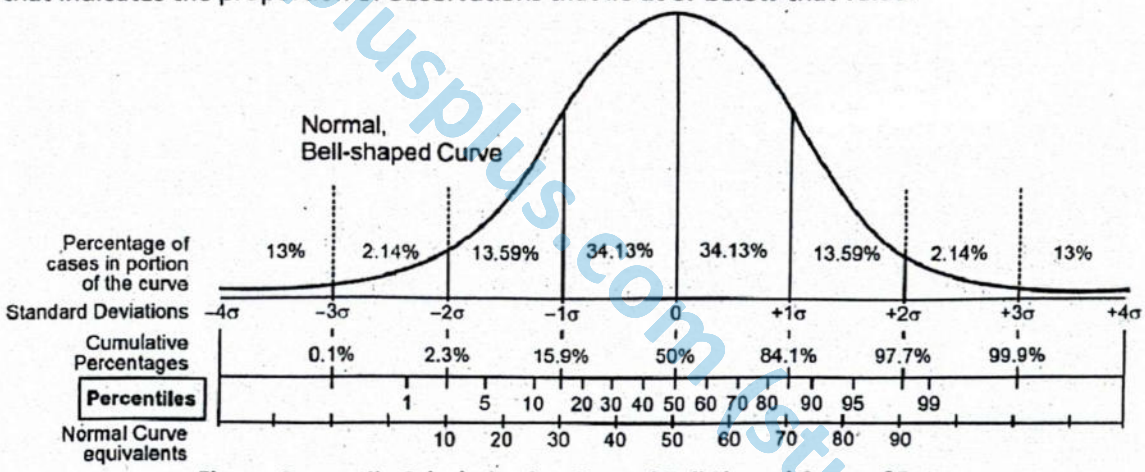


Figure: Percentile Calculation in a Normal Bell Shaped Curve of Data

Further Helping Material to Understand Percentile

Quartiles:

Interquartile Range and Its Significance

The interquartile range (IQR) is defined as the difference between the third quartile (Q_3) and the first quartile (Q_1). It represents the range within which the central 50% of data values lie, and is often used to describe the spread or dispersion of data, excluding extreme values or outliers. In biological datasets, the IQR provides a robust summary of variation, particularly when dealing with non-normally distributed data.

Application of Quartiles and Percentiles in Biological Research

Consider the following example involving the **number of fruits on orange trees in a small garden.** If the **25th percentile** corresponds to a count of **39 fruits**, this means that **25% of the trees produced fewer than 39 fruits.** This allows the researcher to understand the **distribution of fruit production** and identify underperforming trees within the population.

The **50th percentile** in this dataset would represent the **median fruit count**, dividing the population into two equal halves. A tree that falls at the **99th percentile** in fruit production would have yielded **more fruit than 99% of the other trees**, indicating exceptionally high productivity. This type of ranking allows for meaningful comparisons among individuals or subgroups within the overall population.

Use of Percentile Ranks in Biometry

In **biometry and educational testing, percentile ranks** are commonly used to **interpret individual values** in the context of a broader population. For instance, if a student's height is measured at **77 cm**, and this measurement falls into the **89th percentile**, it implies that **89% of the individuals in that population have a height of 77 cm or less.** This comparative analysis enables researchers and educators to evaluate where an individual stands in relation to peers, thereby facilitating **data-driven conclusions** about growth, health status, or academic performance.

Quartile	Number of fruits on each plant
	11
	13
	14
First Quartile	**15**
	18
	18
	19
Second Quartile=(21)	**19**
	23
	23
	23
	24
Third Quartile	**24**
	25
	25
	26

Steps to Calculate Percentile

1. **Arrange Data:** Sort the data values in ascending order.
2. **Calculate Position:** Use the formula for the position of the P^{th} percentile:

$$\text{Position} = \frac{P}{100} \text{X} (N+1)$$

P is the desired percentile (e.g., 25 for the 25th percentile).
N is the number of observations.
Percentile can be calculated by using the data by formula:

$$P = \frac{n}{N} \times 100\%$$

Where P is the percentile, n is the number of data points below the data point of interest, and N is the total number of data points in the data set.

SLO [B-12-K-04]
Sketch a bar chart for a given set of data.

BAR CHART

In biostatistics and scientific research, *data is typically documented in the form of* **numerical figures** and often organized into **tables**. However, large volumes of numerical data, especially when presented in tabular format, may appear **uninspiring and difficult to interpret**. The sheer quantity of figures can make it challenging for researchers and observers to **grasp key patterns or derive meaningful conclusions** at a glance.

To address this limitation, data is frequently represented using **graphs, diagrams, charts, and pictorial illustrations**. These visual tools offer a **more engaging and accessible means** of communicating complex datasets. Unlike tables filled with figures, **graphic presentations provide a clear, concise, and impactful view** of the data, helping to convey the overall message effectively. Visual formats such as **bar charts, rectangular diagrams, pie charts**, and others leave a **long-lasting impression** on the mind of the observer and are particularly useful in educational, medical, and research settings.

Simple bar diagram or simple bar chart

A **simple bar chart** is one of the most commonly used methods for representing **single-variable data**. It consists of a series of **bars of equal width**, where **each bar represents a single value** or figure. These charts are **not subdivided** to represent additional characteristics or classifications within the dataset.

In a simple bar chart:

- Each bar corresponds to **only one characteristic** of the data.
- The **heights (or lengths)** of the bars represent the **magnitude** of the data values.
- While the width of all bars remains **constant**, their **height varies proportionally** according to the quantity or degree of the variable being represented.
- A simple bar chart may contain **multiple bars**, with each bar standing for a different data value or category.

> **SLO [B-12-K-06]**
> **Evaluate the appropriate type of figure or chart for a given set of data and/or experiment (bar chart, pie chart, xy axis data figure etc).**

SELECTING AND EVALUATING FIGURES AND CHARTS

In biological research, data visualization is a powerful tool for summarizing **results, identifying patterns, and communicating findings effectively**. The choice of a figure or chart depends on the type of data, the objective of the analysis, and the audience. Selecting an inappropriate chart can misrepresent **data, obscure trends, or confuse the reader**.

Types of Figures

1. Bar Charts

- **Use:** To compare discrete categories or groups.
- **Structure:** Rectangular bars of equal width, with height representing the value.
- **Example:** Comparing the average number of seeds produced by three different plant species.
- **Why Appropriate:** Shows clear comparisons between categories and allows easy visualization of differences.

2. Pie Charts

- **Use:** To represent parts of a whole as percentages or proportions.
- **Structure:** A circle divided into sectors, with each sector representing a category's contribution to the total.
- **Example:** Distribution of blood types (A, B, AB, O) in a population.
- **Why Appropriate:** Clearly shows proportional contributions but is not suitable for comparing multiple datasets.

3. Line Graphs / XY Scatter Plots

- **Use:** To show relationships, trends, or correlations between two continuous variables.
- **Structure:** Points plotted on an XY coordinate plane, often connected by a line for trends.
- **Example:** Plotting temperature vs. rate of photosynthesis in a plant species.
- **Why Appropriate:** Displays trends over time or continuous changes, allowing evaluation of relationships between variables.

4. **Histograms**

- **Use:** To display the distribution of numerical data in intervals (bins).
- **Structure:** Similar to bar charts but represents frequency of ranges rather than discrete categories.
- **Example:** Distribution of heights of 100 seedlings measured in a nursery.
- **Why Appropriate:** Useful for understanding frequency distributions and identifying patterns like skewness or normality.

5. **Box Plots (Whisker Plots)**

- **Use:** To summarize data distribution, highlighting the median, quartiles, and outliers.
- **Example:** Comparing test scores of students in two different biology classes.
- **Why Appropriate:** Shows spread, central tendency, and potential outliers clearly, especially for comparing multiple datasets.

Choosing the Right Figure

Selecting the most appropriate figure or chart is crucial for accurately representing data and effectively communicating results. A poorly chosen figure can obscure patterns, mislead the audience, or make interpretation difficult. To ensure clarity and accuracy, consider the following steps:

1. **Identify the Type of Data**

Understanding the nature of your data is the first step:

- **Categorical Data:** Represents distinct groups or categories, such as blood types (A, B, AB, O) or plant species. These are best displayed using **bar charts or pie charts.**
- **Numerical Data:** Represents measurable quantities, such as height, weight, or enzyme activity. This can be **discrete** (countable, e.g., number of offspring) or **continuous** (measured on a scale, e.g., temperature). Distributions of numerical data are often shown using **histograms, box plots, or scatter plots.**

2. **Determine the Purpose of the Figure**

Consider what you want the figure to convey:

- **Comparing Categories:** Use bar charts to highlight differences between groups, such as comparing the growth rate of three plant species under different fertilizers.
- **Showing Proportions:** Use pie charts to visualize relative contributions, like the percentage of different cell types in a tissue sample.
- **Illustrating Trends or Relationships:** Use line graphs or XY scatter plots to show changes over time or correlations between variables, such as plotting temperature versus enzyme activity.
- **Analyzing Distributions:** Use histograms or box plots to show how data are spread, identify outliers, or detect skewness in a dataset.

3. **Consider Clarity and Simplicity**

A figure should make patterns immediately apparent without overwhelming the viewer:

- Keep labels clear and legible.
- Avoid excessive colors or decorative elements that distract from the data.
- Include a concise title and units of measurement where relevant.
- **Example:** A bar chart comparing leaf length across four plant species should clearly label each species on the X-axis and the measurement on the Y-axis, avoiding 3D effects that distort perception.

4. **Avoid Misleading Representations**

Accuracy is critical in data visualization. Common pitfalls include:

- **Truncated axes:** Cutting off the axis scale can exaggerate differences between bars or points.
- **Inappropriate scales:** Using logarithmic scales without explanation may confuse readers.
- **Overcomplicated visuals:** Including too many categories or variables in one figure can obscure trends.
- Example: A bar chart showing population growth should start the Y-axis at zero to prevent overstating differences between groups.

5. **Match the Figure to the Audience**

Consider the knowledge level of your audience:

- Scientific peers may appreciate detailed scatter plots with regression lines.
- General audiences benefit more from simplified bar charts, pie charts, or line graphs that focus on the key message.

By carefully assessing the **type of data, purpose, clarity, accuracy, and audience**, you can select figures that not only convey information effectively but also support valid scientific conclusions.

SLO [B-12-K-07]
Make the appropriate chart with proper title, labeled axes, legend, axes units.

SKETCHING OR CONSTRUCTION OF BAR CHARTS AND PIE CHARTS WITH PROPER TITLE, LABELLED AXES, LEGEND, AXES UNITS

With Proper Title, Labelled Axes, Legend, and Axis Units

To construct a scientifically accurate and visually effective bar chart or pie chart, the following systematic steps should be followed:

1. **Select a graph paper** of suitable size and determine the **scale** based on the type and range of data to be represented.
2. **Draw two perpendicular axes** on the graph that intersect at the origin (zero point):
   - The **horizontal line (X-axis)** usually represents the categories or **data items**.
   - The **vertical line (Y-axis)** represents the quantitative values.
3. **Label both axes appropriately**:
   - The X-axis should be marked with the **names of the data items or variables**.
   - The Y-axis should be labelled with the **unit of measurement** corresponding to the data values.
4. **Choose a uniform width** for each bar and maintain **equal spacing** between the bars along the horizontal axis to ensure visual balance.
5. **Determine the scale** on the Y-axis based on the range of data. This scale will be used to calculate and draw the **heights of the bars** accurately.
6. **Mark the heights** of the bars according to the scale using **dots or guidelines.**
7. **Draw the bars carefully**, ensuring each one is proportional to the data it represents. It is important to recheck all measurements and values for accuracy.
8. **Add a descriptive title** to the graph that reflects the purpose or nature of the data being represented.
9. If necessary, include a **legend** to indicate different variables, especially in compound or comparative bar charts or when using pie charts.

► **Some basic steps to make a pie chart manually, follow these steps**

1. **Gather Your Data:** You'll need a list of categories and their values.
   Example: Category A: 20, Category B: 30, Category C: 50
2. **Calculate Total and Percentages:** Add up the values to get the total. Then, for each category, divide its value by the total and multiply by 100 to get its percentage.
   Example: Total = 20 + 30 + 50 = 100, Category A: (20/100) × 100 = 20%, Category B: (30/100) × 100 = 30%, Category C: (50/100) × 100 = 50%
3. **Convert Percentages to Degrees:** Since a circle is 360 degrees, multiply each percentage by 3.6 (because 100% × 3.6 = 360°) to get each angle.
   Example: Category A: 20% × 3.6 = 72°, Category B: 30% × 3.6 = 108°, Category C: 50% × 3.6 = 180°
4. **Draw a Circle:** Use a compass to draw a circle on your paper. Mark the center.
5. **Measure and Draw Each Segment:** Start at the top of the circle (12 o'clock position). Use a protractor to measure each angle from the center, marking each section. Draw lines from the center to the edge of the circle at each angle to divide the circle into sections.

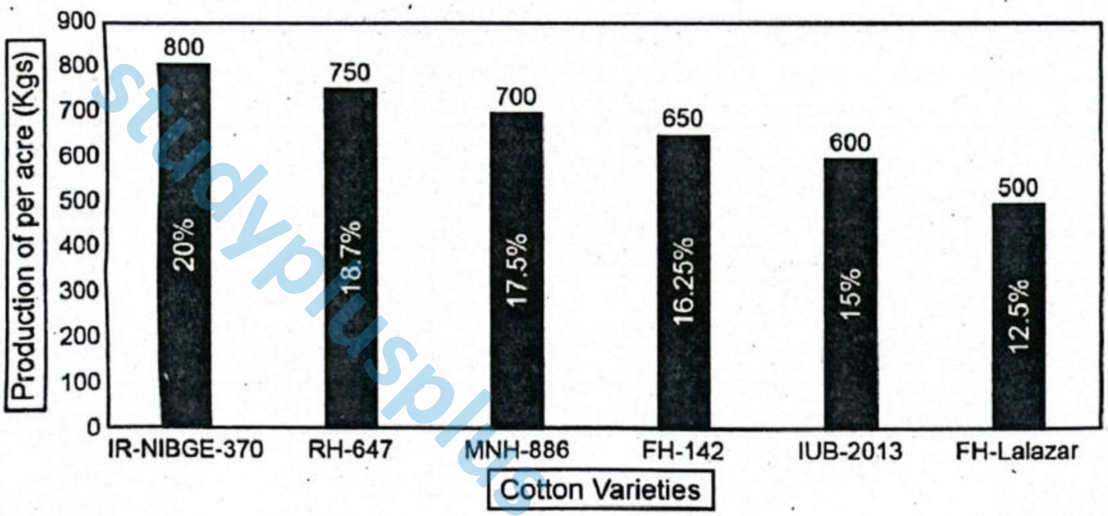
6. **Label Each Segment:** Write each category name and percentage inside or near each section.
7. **Add Color (Optional):** You can color each segment differently for clarity.

Bar chart and pie chart can be constructed by using computer software.

Example 1: Draw simple bar graph/chart to represent the cotton (weight in Kgs) of produced per acer of land from five Bt cotton varieties (IR-NIBGE-370, RH-647, MNH-886, FH-142, IUB-2013 and FH-Lalazar) in year 2023.

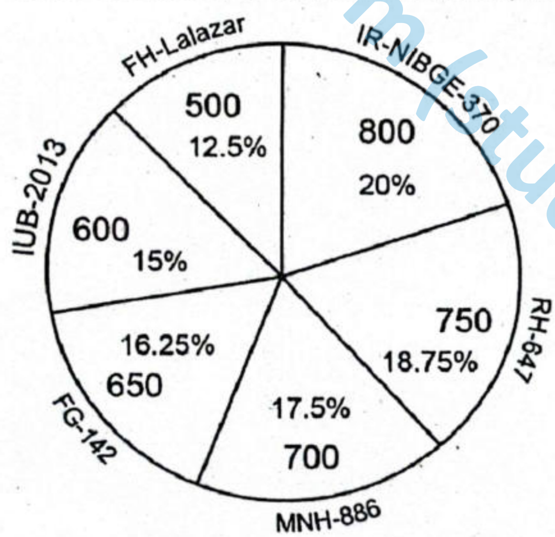
Cotton Varieties	IR-NIBGE370	RH-647	MNH-886	FH-142	IUB-2013	FH-Lalazar
Production per acre (Kgs)	800	750	700	650	600	500
Production per acre (%age)	20%	18.75%	17.5%	16.25%	15%	12.5%

- **Simple bar chart using above data**



- **Simple pie chart using above data**

Pie chart to show Cotton Production of Bt-cotton Varieties per Acre (Kgs) in 2023



- **Trend Line and it's purpose**

A trend line shows the overall direction or pattern in the data over time or across categories. In bar graphs, trend lines are usually drawn as a dotted line across the tops of the bars to show increasing, decreasing or constant trends. Trend line explains that

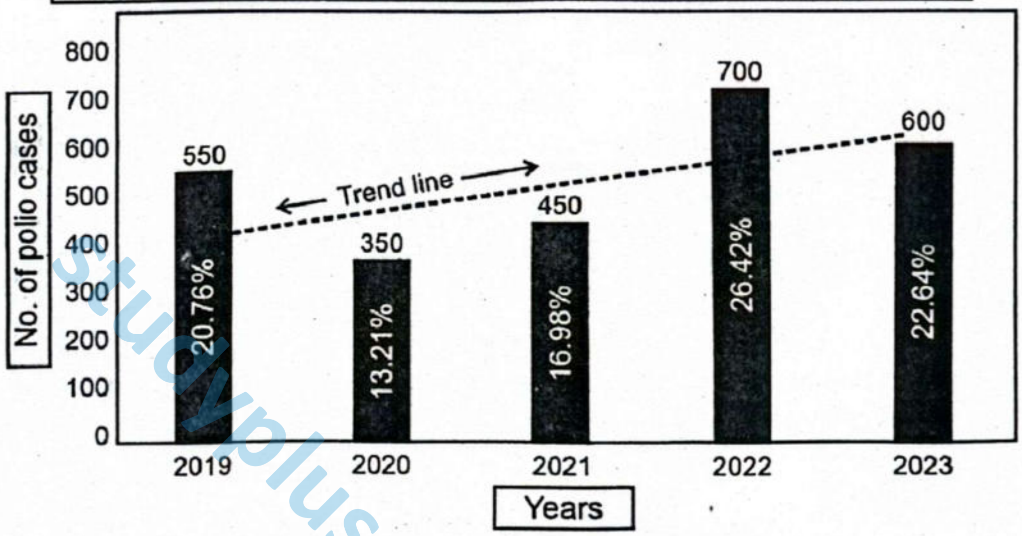
"There is an increasing trend...

"'The trend line shows a gradual decline...

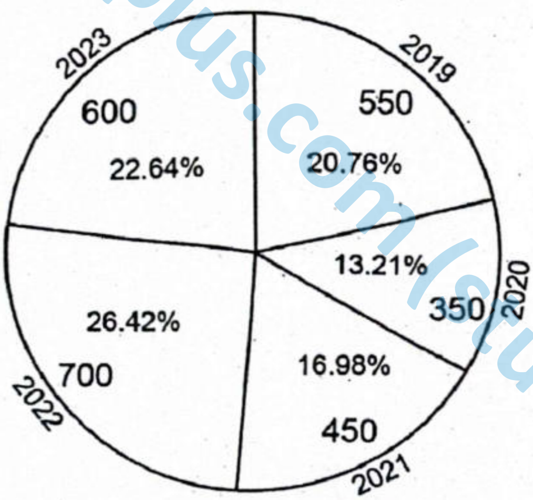
""The trend remains fairly constant...
"'There is a sharp rise/ drop between..." etc.

Example 2: Draw simple bar graph/chart to represent the prevalence of polio, showing number of confirm polio cases in Pakistan in last five years (2019-2023)

Years	2019	2020	2021	2022	2023
No. of polio cases	550	350	450	700	600
%age of polio cases	20.76%	13.21%	16.98%	26.42%	22.64%



Pie chart for Prevalence of Polio in Pakistan in last five years



SLO [B-12-K-06]
Sketch error bars based off of range or standard deviation for a given set of data on a bar chart.

SKETCHING ERROR BARS BASED ON RANGE OR STANDARD DEVIATION ON A BAR CHART

Error bars are graphical elements added to bar charts to represent the **variability** or **uncertainty** in data. They provide insight into the **distribution spread**, helping readers to better understand how much individual values deviate from the mean or another measure of central tendency. Commonly, error bars are used to show **range, standard deviation (SD),** or **standard error (SE).**

1. **Calculate Error Values**

Range-Based Error Bars: For each data point or category, first determine the minimum and maximum values to get the range. The error bar's length will be half of the range, extending above and below the mean or median value.

Standard Deviation (SD) Error Bars: For each data point or category, calculate the standard deviation. Error bars will extend above and below the mean by one standard deviation.

▶ **Range-Based Error Bars**

For each data point or category:

- Identify the minimum and maximum values.
- Calculate the range by subtracting the minimum from the maximum value.
- The length of the error bar is typically half of the range, extending equally above and below the central value (e.g., mean or median).

▶ **Standard Deviation (SD) Error Bars**

For each data point or category:

- Compute the **mean** of the dataset.
- Determine the **standard deviation** to quantify the average dispersion of values.
- Draw error bars that extend **one standard deviation above and below** the mean value.

Note: In scientific publications, the number of standard deviations used (e.g., ±1 SD, ±2 SD) should be clearly specified.

2. **Draw the Bar Chart**

- Construct a **bar chart** in which each bar represents the **mean** (or median) value of a category.
- Use **equal-width bars** with consistent spacing to maintain clarity and visual balance.

3. **Add Error Bars**

▶ **Vertical Error Bars**

- At the **center of the top edge** of each bar, draw a **vertical line** extending **upward and downward** by the calculated error value (range/2 or ±SD).
- Ensure that each error bar line is centered accurately on the bar to reflect symmetrical variability.

▶ **End Caps**

- Add **horizontal caps** at the top and bottom of each error bar line. These caps enhance visibility and provide a **clear, professional appearance**.

▶ **Length of Error Bars**

- The **total length** of each error bar should reflect the **magnitude of the variability**:
  - For range: half of the total range (above and below the central value).
  - For SD: one standard deviation above and below the mean.

Error bar Length:

For range, set the error bar to $\frac{(\text{Max value} - \text{Min value})}{2}$

For standard deviation, set it equal to the standard deviation.

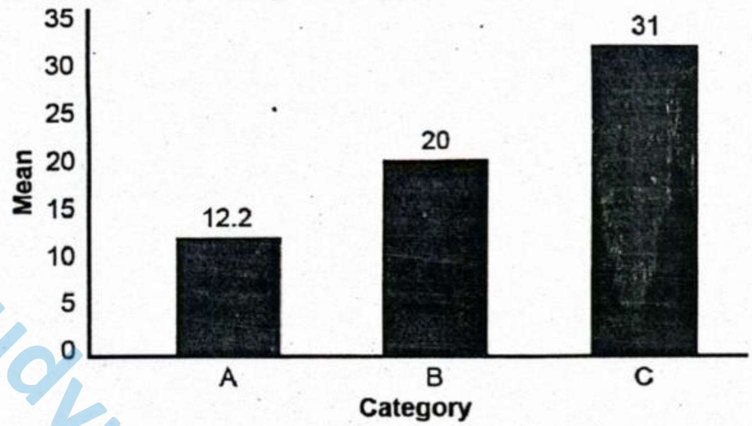
Calculation for Standard Deviation

Suppose we have the following data points for three categories in a biological experiment:

1. Calculate the mean and standard deviation for each category.
2. For each bar representing the mean of each category, add error bars with lengths based on the calculated standard deviations.
3. Here are the data points, mean values, ranges and standard deviation (SD) and error bar lengths for each category:

Category	Data Points	Mean	Range $\frac{(\text{Max value} - \text{Min value})}{2}$	Standard deviation (SD)	lengths of error bars (± SD)
A	(10,12,15,11,13)	12.2	2.5	1.92	±1.92
B	(20,21,22,19,18)	20.0	2	1.58	±1.58
C	(30, 33, 32, 31,29)	31.0	2	1.58	±1.58

Graph showing error bars



SLO [B-12-K-08]
Design an appropriate experiment with control group and dependent, independent and control variables.

EXPERIMENTAL DESIGN WITH CONTROL GROUP AND DEPENDENT, INDEPENDENT AND CONTROL VARIABLES

In scientific experimentation, **two primary types of groups** are utilized to achieve reliable, interpretable, and comparative results: the **experimental group** and the **control group**. These groups serve as the foundation for analyzing the effects of experimental treatments and validating the findings.

- **Experimental Group**

  In scientific experimentation, **two primary types of groups** are utilized to achieve reliable, interpretable, and comparative results: the **experimental group** and the **control group**. These groups serve as the foundation for analyzing the effects of experimental treatments and validating the findings.

- **Control Group**

  The **control group** is maintained under **standard or unchanged conditions**, without receiving the specific treatment applied to the experimental group. This group serves as a **reference or baseline** to evaluate the impact of the independent variable. The control group enables researchers to determine whether the observed effects are truly due to the treatment or simply the result of natural variations.
  Example: In the same fertilizer experiment, the plants that are **not given fertilizer** form the **control group**. These plants help determine whether the growth observed in the experimental group is due to the fertilizer or other environmental factors.

Independent, Dependent and Controlled Variables

There are three main types of variables in scientific investigations: independent, dependent, and controlled variables.

- **Independent Variables**

  The **independent variable** is the variable that the researcher **deliberately changes or manipulates** during the experiment. It is hypothesized to **cause an effect** on another variable and is therefore also referred to as the **input variable** or **causal factor**.

Example: In the fertilizer experiment, the **presence or absence of fertilizer** is the **independent variable**, as it is being changed to study its impact on plant growth.

Note: Only one independent *variable* should be manipulated at a time to ensure that any observed effect in the dependent variable is solely due to that change.

- **Dependent variables**

  The **dependent variable** is the variable that is **measured or observed** during the experiment. It reflects the **outcome** that results from the manipulation of the independent variable and is also known as the **response variable** or **effect variable**.

  Example: In the fertilizer experiment, the **dependent variable** could be **plant height, number of leaves**, or **overall growth rate**—all of which are expected to change based on the fertilizer treatment.

  It is important that dependent variables are **measurable** and ideally **quantitative**, to enable accurate recording, analysis, and comparison.

- **Controlled variables**

  Controlled variables, also known as **constants**, are all other factors that must be **kept the same** in both the experimental and control groups. By maintaining these variables constant, the researcher ensures that **any change in the dependent variable is due solely to the independent variable**, and not to other uncontrolled factors.

  Example: In the fertilizer experiment, **amount of sunlight, temperature, water supply, type of plant**, and **soil quality** are examples of **controlled variables**. If any of these were to change between groups, it could confound the results and compromise the validity of the experiment.

Further examples of independent, dependent and control variables:

Let's consider a researcher investigating the effect of **temperature** on the **rate of photosynthesis in plants.**

- **Independent Variable:**
  The temperature is the independent variable, as it is being deliberately altered to observe its effect.
- **Dependent Variable:**
  The rate of photosynthesis (e.g., measured by oxygen output or glucose production) is the dependent variable, as it is expected to change in response to temperature variations.
- **Controlled Variables:**
  To ensure a fair test, variables such as the type of plant, intensity and duration of light, and concentration of carbon dioxide (CO_2) must remain constant throughout the experiment.

Controlling these variables ensures that **any variation in the rate of photosynthesis** can be confidently attributed to changes in temperature.

Why Only One Independent Variable Should Be Used

It is essential to **manipulate only one independent variable at a time.** If multiple variables are changed simultaneously, it becomes **impossible to determine which one is responsible** for the observed effect in the dependent variable. This principle is central to maintaining the **validity and accuracy** of experimental results.

Experimental Design

Let's consider two examples that illustrate how a controlled experiment is designed to study the effect of different factors on plant growth:

- **Experiment 1: Effect of Light Intensity on Plant Growth**

  Objective: To investigate how different light intensities affect the growth of a specific plant (e.g., bean plants).

  Hypothesis: *Plants exposed to higher light intensities will grow more compared to plants under lower light intensities.*

  Materials required: 20 potted bean plants (similar age and size), light source with adjustable intensity, ruler, water and watering can, soil with same composition, thermometers.

Procedure:

1. Divide the 20 potted plants into five groups of four plants each.
2. Place each group under different light intensities (e.g., 100%, 75%, 50% and 25%).
3. Ensure all groups receive the same amount of water and are kept at a constant temperature.
4. Measure the height of the plants every three days continuously for four weeks.
5. Record and analyze the growth data.

Variables:

Independent Variable: Light intensity (100%, 75%, 50%, 25% and 10%)

Dependent Variable: Plant growth; which we could be measured by tracking the height of the plants (in cm) and number of leaves of the plants.

Control Variables: Type of plant, type of soil, amount of water, temperature, duration of light, amount of CO_2, pot size and initial plant size.

Control Group: Plants grown under the normal light intensity (e.g., 100%)

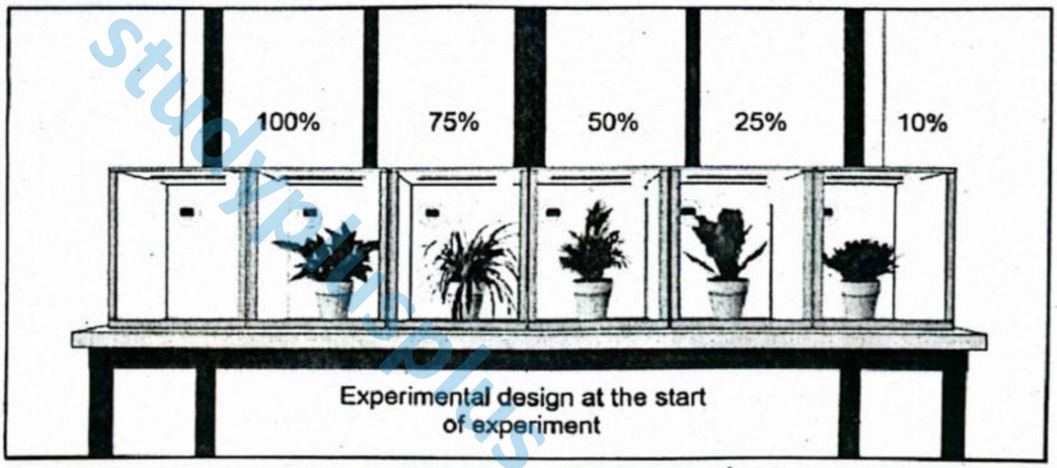


Figure: Experimental design to check the effect of light intensity on plant growth

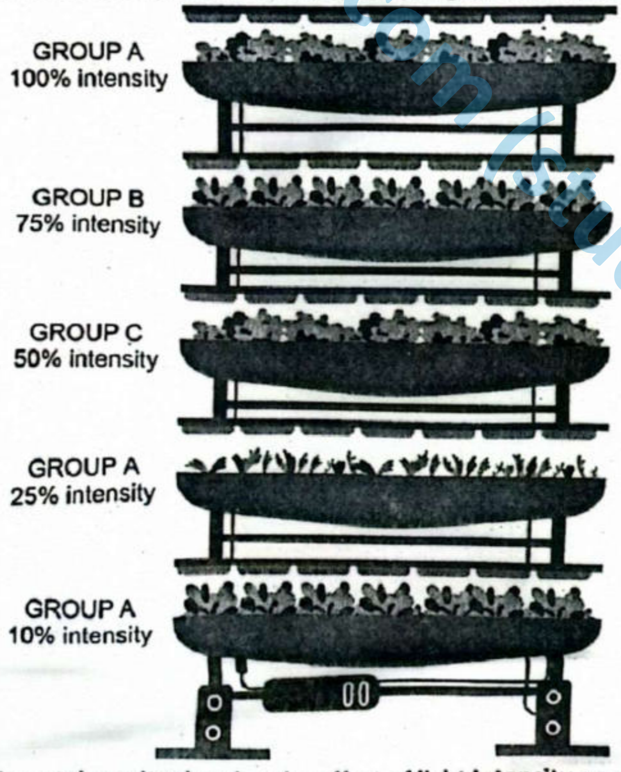


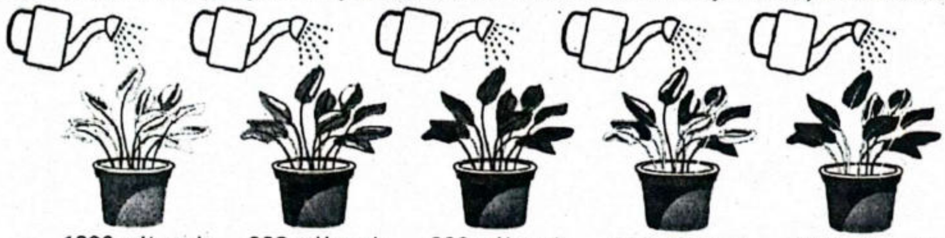
Figure: Experimental results showing the effect of light intensity on plant growth

▶ **Experiment 2: Effect of different amounts of water on plant growth**

Objective: To investigate how different amounts of watering, affects the growth of a specific plant (e.g., potato plants).

Hypothesis: Plants given more amount of water (1000 ml per week) will grow more compared to plants given lower amount of water.

Materials required: 20 potato plants (similar age and size) in lager pots, water and graduated watering can, Light source with same intensity for all plants, ruler, soil with same composition, Thermometer.



Experimental design at the start of experiment

Figure: Experimental design to check the effect of amount of water on plant growth

Procedure:

1. Divide the 20 potato plants potted in larger but same size pots into five groups of four plants each.
2. Place all five groups under light of same higher intensity (100%) and at a constant temperature (30°C).
3. Give different amount of water per week to each group of plants (1000ml, 800ml, 600ml, 400ml and 200 ml).
4. Measure the height (in cm) of the plants and number of leaves on each group of plants after every three days continuously for six weeks.
5. Record and analyze the data of plant growth.

Figure: Experimental results showing the effect of amount of water on plant growth

Variables:

Independent Variable: Amount of water given per week to each group of plants (1000ml, 800ml, 600ml, 400ml and 200 ml).

Dependent Variable: Plant growth; which we could be measured by tracking the height of the plants (in cm) and number of leaves of the plants.

Control Variables: Type and species of plant, type of soil, duration and intensity of light, temperature, amount of CO_2, pot size and initial plant size.

Control Group: Plants watered 1000 ml per week.

▶ **Experiment 3: Effect of Different pH Levels on Enzyme Activity**

Objective: To examine how pH levels affect the activity of the enzyme catalase in breaking down hydrogen peroxide.

Hypothesis: Catalase will have the highest activity at a neutral pH and will be less active in acidic or alkaline conditions.

Materials required: Hydrogen peroxide solution, catalase (can use potato or liver tissue as a source), Test tubes, pH buffer solutions (pH 4, pH 7, pH 10), Stopwatch, Measuring cylinder, Thermometer

Procedure:

1. Place an equal amount of catalase in each test tube.
2. Add 10 mL of hydrogen peroxide to each test tube.
3. Add different pH buffers to each tube to achieve desired specific pH levels for each test tube.
4. Start the stopwatch as soon as hydrogen peroxide is added, and observe the time taken for bubbles to form or the amount of gas produced after 1 minute.
5. Repeat three times for each pH and calculate the average.

Variables:

Independent Variable: pH level (pH 4, pH 7, pH 10).
Dependent Variable: Enzyme activity (rate of reaction measured by gas produced or reaction time).
Control Variables: Amount of hydrogen peroxide, amount of catalase, temperature, and test tube size.
Control Group: Reaction at neutral pH (pH 7).
Above mentioned experiments, will give a hands-on and easily observable understanding to students of scientific methods and the effects of variables on biological processes.

SLOs Based (MCQs)

STANDARD DEVIATION

1. Which of the following best describes standard deviation?
A. The middle value in a dataset
B. The difference between the highest and lowest values
C. The average of all values in a dataset
D. The measure of how spread out values are from the mean

2. If the standard deviation of a dataset is zero, it means that:
A. The values are extremely variable
B. The mean is zero
C. All values are the same
D. There is no data

3. A larger standard deviation in a dataset indicates:
A. Less consistency among the values
B. More similarity among the values
C. A higher mean
D. A lower median

BAR CHART

4. In a simple bar chart, each bar represents:
A. A category's average over time
B. One data value or figure
C. A changing variable over a timeline
D. The frequency of a variable at multiple time points

5. Which of the following is essential when sketching a bar chart?
A. Triangular bars
B. Unequal gaps between bars
C. Labelling axes and providing a title
D. Placing bars diagonally

6. In a bar chart, the height of each bar represents:
A. The size of the sample
B. The measurement scale
C. The magnitude of the data value
D. The range of the dataset

PIE CHART

7. A pie chart is most suitable for representing:
A. Individual scores
B. Time-series data
C. Part-to-whole relationships
D. Dispersion of data

8. In a pie chart, the whole circle represents:
A. A frequency table
B. 50% of the data
C. 100% of the total data
D. Only categorical variables

9. Which of the following is true about pie charts?
A. They are used to display ranges
B. They use bars to display values
C. Each slice is proportional to the category's percentage
D. They are based on continuous data

PERCENTILE

10. The 75th percentile is the value:
A. At the lower end of the dataset
B. Below which 75% of the data lies
C. That equals the mean
D. Above which 75% of values are found

11. If a student's score is in the 90th percentile, it means:
A. 10% of students scored more
B. 90% of students scored higher
C. 10% of students scored the same
D. The student scored below average

12. Which of the following percentiles corresponds to the median?
A. 25^{th} B. 75^{th}
C. 50^{th} D. 99^{th}

RANGE

13. Range is calculated by:
A. *Dividing total values by number of values*
B. Subtracting the lowest value from the highest
C. Adding the highest and lowest values
D. Subtracting mean from each value

14. Which of the following is a limitation of using range?
A. It considers all data points equally
B. It is affected by outliers
C. It is difficult to calculate
D. It gives a precise measure of central tendency

15. A small range in a biological experiment indicates:
A. High variation in responses
B. Greater effect of outliers
C. Low consistency in results
D. Uniformity among subjects

MEDIAN

16. Median is defined as:
A. The highest value in the dataset
B. The most frequently occurring value
C. The middle value when data is arranged in order
D. The average of all data points

17. Which type of data is most suitable for using the median as a measure of central tendency?
A. Data with no variation
B. Data with extreme outliers
C. Uniform data
D. Data that is normally distributed

CONTROL AND EXPERIMENTAL GROUPS

18. The group in an experiment that receives the treatment is called the:
A. Control group B. Test group
C. Experimental group D. Reference group

19. The control group in an experiment is used to:
A. Ensure that the sample size is increased
B. Increase the variability of the data
C. Compare results with the experimental group
D. Eliminate the need for statistical analysis

20. Which of the following statements about experimental design is correct?
A. Only the dependent variable is changed
B. The experimental group serves *as a benchmark*
C. Controlled variables should change during the experiment
D. Only the independent variable is deliberately manipulated

Answer Key

1. D	2. C	3. A	4. B	5. C	6. C	7. C	8. C	9. C	10. B
11. A	12. C	13. B	14. B	15. D	16. C	17. B	18. C	19. C	20. D

Numerical Problems

1. Find the mean of the following data: 12, 18, 25, 30, 15.
Step 1: Add all the numbers: 12 + 18 + 25 + 30 + 15 = 100
Step 2: Count the number of values: 5
Step 3: Divide the sum by the number of values: 100 ÷ 5 = 20
Answer: Mean = 20

MARKING SCHEME (3 Marks)
- 1 mark for each step

2. Find the median of the following data: 8, 11, 15, 18, 21.
Step 1: Arrange data in ascending order (already arranged): 8, 11, 15, 18, 21
Step 2: Find the middle value: 15 (3rd value)
Answer: Median = 15

MARKING SCHEME (3 Marks)
- *1.5 marks for each step*

3. **Find the mode of the following data: 5, 8, 9, 8, 10, 11, 8.**
Step 1: Count the frequency of each number:
5: 1 time, 8: 3 times, 9: 1 time, 10: 1 time, 11: 1 time
Step 2: The most frequent value is 8
Answer: Mode = 8

MARKING SCHEME (3 Marks)
- 1.5 marks for each step

4. **Calculate the range of the following values: 45, 60, 30, 25, 55.**
Step 1: Identify maximum value = 60
Step 2: Identify minimum value = 25
Step 3: Range = 60 – 25 = 35
Answer: Range = 35

MARKING SCHEME (3 Marks)
- 1 mark for each step

5. **Find the standard deviation (population) of: 5, 7, 9.**
Step 1: Find the mean = (5 + 7 + 9)/3 = 21/3 = 7
Step 2: Find squared deviations: $(5-7)^2 = 4$, $(7-7)^2 = 0$, $(9-7)^2 = 4$
Step 3: Mean of squared deviations = (4 + 0 + 4)/3 = 8/3 ≈ 2.67
Step 4: Standard deviation = $\sqrt{2.67} \approx 1.63$
Answer: SD ≈ 1.63

MARKING SCHEME (3 Marks)
- 0.75 mark for each step

6. **Find the mean of: 40, 50, 60, 70, 80, 90.**
Step 1: Add all numbers: 40 + 50 + 60 + 70 + 80 + 90 = 390
Step 2: Count of numbers = 6
Step 3: Mean = 390 ÷ 6 = 65
Answer: Mean = 65

MARKING SCHEME (3 Marks)
- 1 mark for each step

7. **Calculate the median of: 14, 22, 18, 20, 19.**
Step 1: Arrange in ascending order: 14, 18, 19, 20, 22
Step 2: Middle value = 19 (3rd value)
Answer: Median = 19

MARKING SCHEME (3 Marks)
- 1.5 marks for each step

8. **Calculate the mode for the dataset: 2, 3, 2, 4, 5, 3, 3.**
Step 1: Frequency count: 2: 2 times, 3: 3 times, 4: 1 time, 5: 1 time
Step 2: The most frequent value is 3
Answer: Mode = 3

MARKING SCHEME (3 Marks)
- 1.5 marks for each step

9. **Calculate the range for the data: 101, 97, 105, 110, 99.**
Step 1: Max = 110, Min = 97
Step 2: Range = 110 – 97 = 13
Answer: Range = 13

MARKING SCHEME (3 Marks)
- 1.5 marks for each step

10. **Calculate population standard deviation for: 2, 4, 6, 8.**
Step 1: Mean = (2 + 4 + 6 + 8)/4 = 20/4 = 5
Step 2: Squared deviations: $(2-5)^2 = 9$, $(4-5)^2 = 1$, $(6-5)^2 = 1$, $(8-5)^2 = 9$
Step 3: Mean of squared deviations = (9+1+1+9)/4 = 20/4 = 5
Step 4: Standard deviation = $\sqrt{5} \approx 2.24$
Answer: SD ≈ 2.24

MARKING SCHEME (3 Marks)
- 0.75 mark for each step

Exercise

I Multiple Choice Questions (MCQs)

❖ **Select the correct Answer.**

1. **In experiment examining the effects of pH levels on the activity of the enzyme, independent variable will be**
A. pH B. Temperature
C. Substrate D. Reaction time

2. **The variable that the experimenter changes in the experiment to check its effect during experiment is called.**
A. Dependent variable B. Independent variable
C. Control variable D. Experimental variable

3. **Variables that researcher measure to determine the effect of the independent variable is also known as:**
A. Outcome variables B. Input variables
C. Cause variables D. Control variable

4. **To which of the following fields of life sciences, biostatistics is not applicable?**
A. Health B. Chemical engineering
C. Agriculture D. Medicine

5. **25^{th} percentile is the value of the data points lie.**
A. Above which 25% B. Below which 25%
C. $3/4^{th}$ D. 25

6. **Biostatistics is not involved in:**
A. Epidemiological studies B. Agriculture
C. Petroleum industry D. Public Health

7. **The factors that must be kept constant during an experiment across both the experimental and control groups are known as**
A. Control variables B. Outcome variables
C. Input variables D. Cause variables

8. **All of the followings are types of averages except:**
A. Arithmetic Mean B. Median
C. Mode D. Product of data values

9. **The symbol for mean is:**
A. $\overline{x}$ (read as "x bar")
B. $\tilde{x}$ (read as "x tilde")
C. $\hat{x}$ (read as "x-hat")
D. h ("read as highly stable")

10. **Formula of "Mean" for un-group data is:**
A. $\frac{\sum x}{n}$ B. $\frac{\Sigma fx}{\Sigma f}$
C. $\frac{\Sigma f1}{\Sigma f2}$ D. $\frac{n+1}{2}$

11. **Find the mode from this dataset given 46,49,55,58,62,66,64,65,66,67,68,70,72:**
A. 46 B. 62
C. 66 D. 72

30
30

12. **The first step in genetic engineering is:**
A. Isolation of the gene of interest
B. Insertion of the gene into a vector
C. Transfer of recombinant DNA into the host organism
D. Expression of the gene

13. **Biostatistics is similar to maths because of:**
A. Calculations
B. Counting birds
C. Used of symbol pie
D. It is also useful in engineering studies

14. **If the numbers of diseased persons is greater in 2nd year of study than 1st year then**
A. Height of the bar will be greater in 1^{st} year
B. Width of the bar will be greater in 1^{st} year
C. Height of the bar will be smaller in 1^{st} year
D. Width of the bar will be greater in 2^{nd} year

15. **In a bar chart of number of deaths due to hepatitis in Pakistan in last 10 years what will we write at the x-axes of graph?**
A. Years B. Number of deaths
C. Any of the value D. Causes of hepatitis

Answer Key

1. A	2. B	3. A	4. B	5. B	6. C	7. A	8. D	9. A	10. A	11. C	12. A	13. A	14. C	15. A

II Short Answer Questions

Q.1 How is Standard Deviation different from Range?

Ans. Standard deviation measures the average spread of values around the mean, showing overall data variability.
Range only shows the difference between the highest and lowest values.
Thus, standard deviation gives a more detailed view of dispersion than range.

Q.2 What are the demerits of relying on range to decide future plans in scientific experiments?

Ans. Range is influenced by extreme values and ignores how data is distributed in between.
It does not reflect consistency or clustering of values.
Relying on range alone can lead to misleading conclusions in experiments.

Q.3 Why is the control group important in scientific experiments?

Ans. A control group provides a baseline to compare the effects of a variable.
It helps in determining whether the observed outcome is due to the treatment or another factor.
Control groups enhance the reliability of experimental results.

Q.4 Define mean, median, and mode.

Ans.
- **Mean:** The mean is the arithmetic average of a dataset, calculated by dividing the sum of all values by the total number of values.
- **Median:** The median is the middle value in a dataset arranged in ascending or descending order; if the number of observations is even, it is the average of the two central values.
- **Mode:** The mode is the value that occurs most frequently in a dataset; a dataset may have one mode (unimodal), more than one mode (bimodal or multimodal), or no mode at all.

Q.5 Write down the formulae of median and mode.

Ans.
- **Median** (for odd n) = Middle value
- **Median** (for even n) = (n/2th value + (n/2 + 1)th value) / 2
- **Mode** (for grouped data) = $L + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$

Where:

L = lower boundary,
f_1 = modal class frequency,
f_0 = preceding class frequency,
f_2 = following class frequency,
h = class width

Q.6 What is the importance of mean calculation?

Ans. Mean provides a central value for a dataset, summarising the overall trend.
It is useful in comparing groups and identifying general patterns in data.
Mean is widely used in scientific, economic, and social analyses.

Q.7 Why is the value of mean not completely reliable to get the correct picture?

Ans. Mean is sensitive to extreme values (outliers), which can distort results.
It may not represent skewed or bimodal data accurately.
Hence, median or mode may be better indicators in such cases.

Q.8 Write the formula for calculation of Mode.

Ans. For ungrouped data: Mode = Value with the highest frequency
For grouped data:

$$\textbf{Mode} = L + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$$

Q.9 How is a bar chart different from a pie-chart?

Ans. A bar chart displays data using rectangular bars, with height representing frequency.
A pie chart shows data as slices of a circle, representing proportion.
Bar charts are ideal for comparisons; pie charts are better for showing percentages.

Q.10 Write one similarity and one difference between rectangle graph and pie graph.

Ans. **Similarity:** Both represent categorical data visually.
Difference: Rectangle (bar) graphs use bars to show frequency; pie charts use circle slices to show proportions.

Q.11 How is graphical representation of data more important than tabular representation?

Ans. Graphs make data easier to interpret at a glance and highlight trends, patterns, and outliers.
They *enhance visual* learning and communication.
Tabular data is detailed, but harder to analyse quickly.

Q.12 Find the standard deviation of the values: 33, 55, 66, 84, 91, 72, 68, 87, 78, 66, 82, 46, 71, 31, 38

Given Values:

Set (n = 15):
33, 55, 66, 84, 91, 72, 68, 87, 78, 66, 82, 46, 71, 31, 38
Mean (μ) = 64.73 (already calculated)

Step 1: Subtract the Mean and Square the Result

x	$x - \mu$	$(x - \mu)^2$
33	−31.73	1006.79
55	−9.73	94.67
66	1.27	1.61
84	19.27	371.33
91	26.27	690.11
72	7.27	52.85
68	3.27	10.69
87	22.27	495.95
78	13.27	176.09
66	1.27	1.61
82	17.27	298.25
46	−18.73	350.81
71	6.27	39.31
31	−33.73	1137.71
38	−26.73	714.49

Step 2: Sum of Squared Differences

$\Sigma(x - \mu)^2 = 5442.31$

Step 3: Calculate the Variance

Variance $(\sigma^2) = \Sigma(x - \mu)^2 / n = 5442.31 / 15 = 362.82$

Step 4: Take the Square Root of Variance

Standard Deviation $(\sigma) = \sqrt{362.82} = 19.05$

Q.13 Calculate the median of the dataset in question 12.

Ans. Ordered values: 31, 33, 38, 46, 55, 66, 66, 68, 71, 72, 78, 82, 84, 87, 91

Total values (n) = 15

Median = 68 (8th value in the ordered list)

III Long Answer Questions

Q.1 What are advantages and disadvantages of Median?

Ans. Advantages of Median:

- **Not affected by extreme values:** Unlike the mean, the median is resistant to outliers and skewed data, providing a better measure of central tendency when data is not symmetrically distributed.
- **Useful for ordinal data:** Median can be used when data is ranked or ordered but not measured precisely (e.g. rating scales).
- **Easy to understand and calculate:** Especially with small datasets, it is straightforward to determine the middle value.

Disadvantages of Median:

- **Ignores actual values of data:** It does not consider the magnitude of all values, only their position.
- **Not suitable for further statistical analysis:** Unlike mean, it cannot be used in algebraic calculations or inferential statistics.
- **Sensitive to data arrangement:** Requires data to be sorted, which can be time-consuming for large datasets.

Q.2 Describe detailed procedure to add error bars in the graph.

Ans. Add Error Bars:

Vertical Error Bars: Draw a line centered on each bar, extending upward and downward by the calculated range or standard deviation. Use caps on the ends of each line for a clean representation.

Error bar Length: For range, set the error bar to

$$\frac{(\text{Max value} - \text{Min value})}{2}$$

For standard deviation, set it equal to the standard deviation.

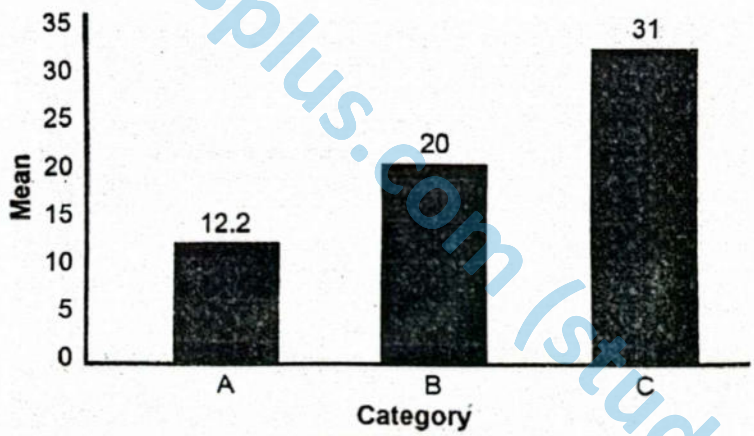
Calculation for Standard Deviation

Suppose we have the following data points for three categories in a biological experiment:

1. Calculate the mean and standard deviation for each category.
2. For each bar representing the mean of each category, add error bars with lengths based on the calculated standard deviations.
3. Here are the data points, mean values, ranges and standard deviation (SD) and error bar lengths for each category:

Category	Data Points	Mean	Range $\frac{(\text{Max value} - \text{Min value})}{2}$	Standard deviation (SD)	lengths of error bars ($\pm$ SD)
A	(10,12,15,11,13)	12.2	2.5	1.92	±1.92
B	(20,21,22,19,18)	20.0	2	1.58	±1.58
C	(30, 33, 32, 31,29)	31.0	2	1.58	±1.58

Graph Showing Error Bars:



Q.3 Complete the following table by filling the values of Cumulative frequency.

Ans.

Class (no. of flowers/plant)	Frequency (no. of plants = n)	Cumulative Frequency
1	30	30
2	50	80
3	60	140
4	40	180
5	30	210

Q.4 Give details steps to construct the Bar-chart.

Ans. Some Basic Steps in Sketching of Bar Charts are:

1. Take a graph paper with suitable sizes and scales as per requirement and type of data.
2. Draw two lines on the graph, perpendicular to each other and intersecting at zero.
3. Label the axis of the graph, horizontal line is X-axis and vertical line is Y-axis.
4. Choose uniform width of bars and uniform gap between the bars along the horizontal axis (X axis).

5. Label X-axis with the names of the data items whose values are to be presented.
6. Choose a suitable scale to determine and specify the heights of the bars for the given values along the vertical axis (Y-axis).
7. Label the Y-axis scale with specific unit of data value to grade the length of bars.
8. Calculate the heights of the bars. According to the scale chosen and mark as dots.
9. Draw the bars carefully and recheck all the entries.

Bars in the graph may be drawn vertically or horizontally but normally vertical bars are used.

Q.5 For what the letters l, f_m, f_1, f_2 and h stands for in the given formula? For which calculation this formula is used? $l+\left[\frac{(f_m-f_1)\times h}{(f_m-f_1)+(f_m-f_2)}\right]$

Ans. This formula is for Mode = the most frequent or repeated value occurring in a given dataset.

Formula for calculating Mode for Group data is given below

Mode ($\hat{X}$) $= l+\left[\frac{(f_m-f_1)\times h}{(f_m-f_1)+(f_m-f_2)}\right]$

Where, l = lowercase boundary of model group

f_m = maximum frequency of a group

f_1 = previous frequency of fm

f_2 = next frequency of fm

h = class interval of model group

Q.6 Sketch and label the Pie-chart by using the data given in the table below about the deaths of vulture birds due to pollution in last five years? Also label the data values in the pie-chart.

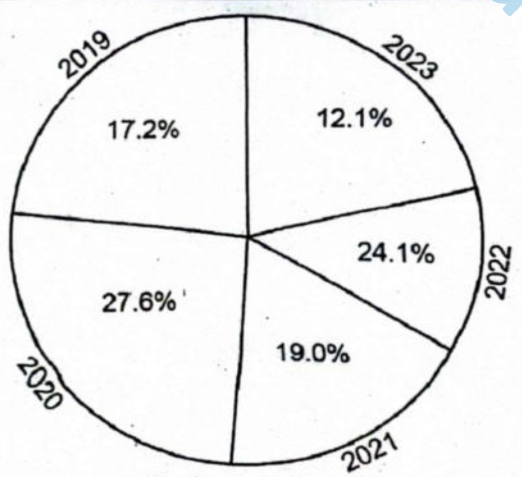
Years	2019	2020	2021	2022	2023
No. of deaths	250	400	275	350	175

Ans. Calculate the Total Number of Deaths:

Total = 250 + 400 + 275 + 350 + 175 = 1450

Convert each Value into an Angle:

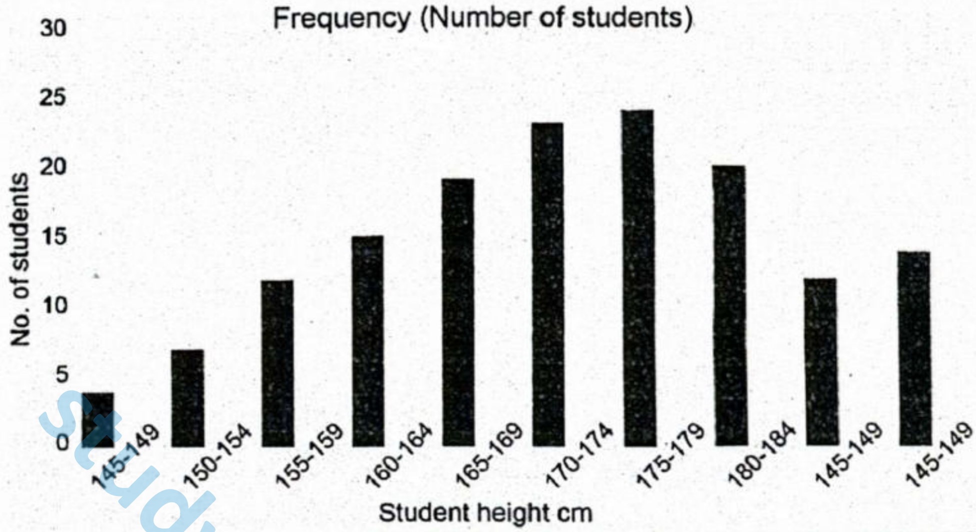
Year	Deaths	Angle (°)	Percentage
2019	250	(250/1450)×360 = 62.07°	17.24%
2020	400	(400/1450)×360 = 99.31°	27.59%
2021	275	(275/1450)×360 = 68.21°	18.97%
2022	350	(350/1450)×360 = 86.90°	24.14%
2023	175	(175/1450)×360 = 43.45°	12.07%



Vulture Deaths due to Pollution (2019-2023)

Q.7 Draw a table and then sketch and label the bar-chart by using the data of height (in cm) collected from 200 students of 12th class of your college.

Ans.



Q.8 Compare and contrast control variables, dependant variables and independent variables.

Ans. In scientific experiments, variables are classified based on their role in the investigation:

- **Independent Variable:** This is the variable that the researcher deliberately changes or manipulates to observe its effect. It is considered the cause in the cause-effect relationship.
- **Dependent Variable:** This is the variable that is measured or observed in response to changes in the independent variable. It represents the effect or outcome of the experiment.
- **Control Variables:** These are the variables that are kept constant throughout the experiment to ensure that the effect on the dependent variable is only due to the independent variable. They eliminate alternate explanations for observed changes.

Comparison:

- The independent variable is manipulated, the dependent variable is measured, and control variables are held constant.
- Independent and dependent variables are related through cause and effect, while control variables ensure the reliability of this relationship.

Q.9 Describe the usage of percentile along with its demerits.

Ans. A percentile is a statistical measure indicating the value below which a given percentage of observations in a group falls. For example, the 60th percentile means 60% of the data falls below that value.

Usage:

- Percentiles are widely used in educational testing, growth charts, and performance evaluations.
- They help in ranking individuals or values relative to a group.
- They are useful in identifying outliers or extreme values in datasets.

Demerits:

- Percentiles do not reflect the exact score difference between individuals (non-linear intervals).
- They can be misleading for small sample sizes.
- They may oversimplify variability and hide patterns within the data.

Q.10 Describe stepwise details to make the appropriate chart with proper title, labeled axes, legend, and axis units from a fictitious data set.

Ans. With Proper Title, Labelled Axes, Legend, and Axis Units:

To construct a scientifically accurate and visually effective bar chart or pie chart, the following systematic steps should be followed:

1. **Select a graph paper** of suitable size and determine the **scale** based on the type and range of data to be represented.
2. **Draw two perpendicular axes** on the graph that intersect at the origin (zero point):
   - The horizontal line (X-axis) usually represents the categories or data items.
   - The vertical line (Y-axis) represents the quantitative values.
3. **Label both axes appropriately:**
   - The X-axis should be marked with the **names of the data items or variables**.
   - The Y-axis should be labelled with the **unit of measurement** corresponding to the data values.
4. **Choose a uniform width** for each bar and maintain **equal spacing** between the bars along the horizontal axis to ensure visual balance.
5. **Determine the scale** on the Y-axis based on the range of data. This scale will be used to calculate and draw the **heights of the bars** accurately.
6. **Mark the heights** of the bars according to the scale using **dots or guidelines.**
7. **Draw the bars carefully**, ensuring each one is proportional to the data it represents. It is important to recheck all measurements and values for accuracy.
8. **Add a descriptive title** to the graph that reflects the purpose or nature of the data being represented.
9. If necessary, include a **legend** to indicate different variables, especially in compound or comparative bar charts or when using pie charts.

Some Basic Steps to MAKE a Pie Chart Manually, Follow these Steps:

1. **Gather Your Data:** You'll need a list of categories and their values.
   Example: Category A: 20, Category B: 30, Category C: 50
2. **Calculate Total and Percentages:** Add up the values to get the total. Then, for each category, divide its value by the total and multiply by 100 to get its percentage.
   Example: Total $=20+30+50=100$, Category A: $(20/100)\times100=20\%$, Category B: $(30/100)\times 100=30\%$, Category C: $(50/100)\times100=50\%$
3. **Convert Percentages to Degrees:** Since a-circle is 360 degrees, multiply each percentage by 3.6 (because $100\%\times3.6=360^{0}$) to get each angle.